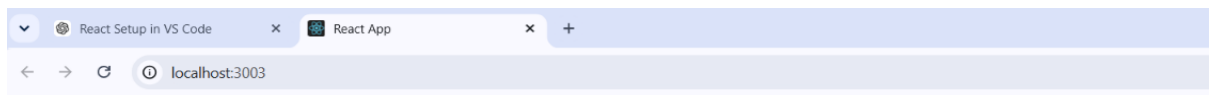
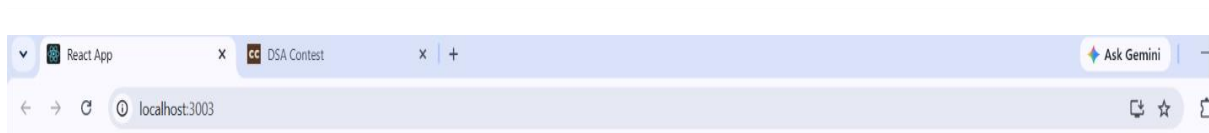


Output19exe19-gitclientappReactJS-HOL



GitHub Repository Viewer

Load Repositories



GitHub Repository Viewer

Load Repositories

- ArrayListDemo
- ArrayListDemo01
- AzureDevopsDemoProductsApi
- CICDMvcDemo
- CleanArchitecture
- ContosoUniversity
- DelagatesDemo
- EventsDemo
- FISGlobal-0721
- GenericCollections
- GenericsDemo
- GitHubDemoApp
- GitTestFolder
- HashTableDemo
- HelloJenkins
- HelloWorldJenkinsAngular
- Layered-Architecture-Demo
- musical-octo-carnival
- mvc03
- Mvvm-With-WPF
- MyApi
- ReflectionDemo
- stunning-broccoli
- techiesyed
- TestRepo
- Webinar-Automation-testing-Selenium-CSharp
- WhyUseGenerics

Objectives

1. Understanding the Need for Isolation in Testing

Isolation in testing means testing a single component or function independently without relying on other parts of the application, such as databases, APIs, or external services. This ensures that the test focuses only on the functionality of the component being tested, making it easier to identify and fix issues.

Why is Isolation Important?

- Detects bugs in individual components quickly.
- Makes tests faster and more reliable.
- Eliminates dependency on external systems.

2. Understanding the Concept of Mocking

Mocking is the process of creating fake versions of functions, modules, APIs, or services during testing. Instead of using the real implementation, a mock object returns predefined data, allowing developers to test application behavior without depending on external resources.

Benefits of Mocking

- Prevents real API or database calls.
- Speeds up test execution.
- Produces predictable and repeatable results.
- Allows testing of error scenarios easily.

3. Using Jest for Unit Testing and Mocking

Jest is a popular JavaScript testing framework developed by Meta (Facebook). It is widely used with React applications to perform unit testing, snapshot testing, and mocking. Jest provides built-in features for creating test cases, verifying expected results, and mocking dependencies.

Key Features of Jest

- Easy to set up with React applications.
- Supports unit testing and integration testing.
- Built-in mocking functionality.
- Provides snapshot testing for React components.
- Generates test coverage reports.
- Runs tests quickly with minimal configuration.